

1. Create user with name Techie and provide sudo access to user

Objective : Create user with name Techie and provide sudo access to user

Purpose : To let them perform administrative tasks without giving them the master password to the entire system

Implementation :

```
root@sunith-virtual-machine:~# su sunith
sunith@sunith-virtual-machine:/root$ cd
sunith@sunith-virtual-machine:~$ su techie
Password:
$ sudo apt install git
[sudo] password for techie:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.34.1-1ubuntu1.17).
git set to manually installed.
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
$
```

```
root@sunith-virtual-machine:~# sudo useradd techie
root@sunith-virtual-machine:~# passwd tcechie
passwd: user 'tcechie' does not exist
root@sunith-virtual-machine:~# passwd techie
New password:
BAD PASSWORD: The password is a palindrome
Retype new password:
passwd: password updated successfully
root@sunith-virtual-machine:~# su techie
$ pwd
/root
$ ^C
$ exit
root@sunith-virtual-machine:~# sudo su techie
$ pwd
/root
$ exit
root@sunith-virtual-machine:~# cd
root@sunith-virtual-machine:~# sudo su sunith
sunith@sunith-virtual-machine:/root$ cd
sunith@sunith-virtual-machine:~$ sudo su techie
[sudo] password for sunith:
$ pwd
/home/sunith
$ sudo apt install git
[sudo] password for techie:
techie is not in the sudoers file. This incident will be reported.
$
```

Conclusion : Successfully implementation is completed

2. Navigate to the home directory.

Objective : Navigate to the home directory

Purpose : Moving our current location in the linux to personal folder

Implementation :

```
sunith@sunith-virtual-machine:~$ sudo su root
root@sunith-virtual-machine:/home/sunith# sudo su sunith
sunith@sunith-virtual-machine:~$ pwd
/home/sunith
sunith@sunith-virtual-machine:~$
```

Conclusion : Successfully Implementation is completed

3. Create a new directory.

Objective : To create a new Folder/Directory

Purpose : Create a new Directory

Implementation :

```
sunith@sunith-virtual-machine:~$ ls
sunith@sunith-virtual-machine:~$ mkdir techie
sunith@sunith-virtual-machine:~$ ls
techie
sunith@sunith-virtual-machine:~$
```

Conclusion : Successfully Implementation Is completed

4. List the contents of a directory.

Objective : List the contents of a directory

Purpose : It displays the contents of a directory in a list format

Implementation :

```
sunith@sunith-virtual-machine:~$ ll /etc/
total 1220
drwxr-xr-x 138 root root 12288 Aug 14 11:42 ./
drwxr-xr-x 20 root root 4096 Aug 10 13:51 ../
drwxr-xr-x 3 root root 4096 Sep 11 2024 acpi/
-rw-r--r-- 1 root root 3028 Sep 11 2024 adduser.conf
drwxr-xr-x 3 root root 4096 Sep 11 2024 alsa/
drwxr-xr-x 2 root root 4096 Aug 13 10:15 alternatives/
-rw-r--r-- 1 root root 335 Mar 23 2022 anacrontab
-rw-r--r-- 1 root root 433 Mar 23 2022 apg.conf
drwxr-xr-x 5 root root 4096 Sep 11 2024 apm/
drwxr-xr-x 3 root root 4096 Aug 12 22:27 apparmor/
drwxr-xr-x 8 root root 4096 Aug 12 22:29 apparmor.d/
drwxr-xr-x 4 root root 4096 Aug 12 22:28 appport/
-rw-r--r-- 1 root root 769 Feb 22 2022 appstream.conf
drwxr-xr-x 8 root root 4096 Aug 10 13:59 apt/
drwxr-xr-x 3 root root 4096 Aug 12 22:28 avahi/
```

Conclusion : Successfully Implementation is Completed

5. Change the current directory.

Objective : Change the current directory

Purpose : Moving from our current folder to another folder

Implementation :

```
sunith@sunith-virtual-machine:~$ ls
techie
sunith@sunith-virtual-machine:~$ cd techie/
sunith@sunith-virtual-machine:~/techie$
```

Conclusion : Successfully Implementation is Completed

6. Create a new empty file.

Objective : Create a new empty file

Purpose : Creating a new file to make a content in that

Implementation :

```
sunith@sunith-virtual-machine:~$ touch file.txt
sunith@sunith-virtual-machine:~$ cat file.txt
sunith@sunith-virtual-machine:~$
```

Conclusion : Successfully Implementation is completed

7. View the contents of a file.

Objective : View the contents of a file

Purpose : Opening a file to read the text inside it right on your scree

Implementation :

```
sunith@sunith-virtual-machine:~$ cat file.txt
this line is added in vim editor
sunith@sunith-virtual-machine:~$
```

Conclusion : Successfully Implementation is Completed

8. Copy a file to another location.

Objective : Copy a file to another location.

Purpose : Making a Duplicate same file in another location

Implementation :

```
sunith@sunith-virtual-machine:~$ cp file.txt /tmp/
sunith@sunith-virtual-machine:~$ ll /tmp/
total 100
drwxrwxrwt 20 root root 4096 Aug 14 12:00 ./
drwxr-xr-x 20 root root 4096 Aug 10 13:51 ../
-rw-rw-r-- 1 sunith sunith 33 Aug 14 12:00 file.txt
drwxrwxrwt 2 root root 4096 Aug 14 11:00 .font-unix/
drwxrwxrwt 2 root root 4096 Aug 14 11:00 .ICE-unix/
```

Conclusion : Successfully Implementation is Completed

9. Move a file to another location.

Objective : Move a file to another location

Purpose : Means shifting a file from its current folder into a completely different folder

Implementation :

```
sunith@sunith-virtual-machine:~$ ls
Screenshots snap suniht.txt
sunith@sunith-virtual-machine:~$ mv suniht.txt /tmp/Horizon.txt
sunith@sunith-virtual-machine:~$ ll /tmp/
total 112
drwxrwxrwt 23 root root 4096 Aug 14 12:14 ./
drwxr-xr-x 20 root root 4096 Aug 10 13:51 ../
-rw-rw-r-- 1 sunith sunith 33 Aug 14 11:54 file.txt
drwxrwxrwt 2 root root 4096 Aug 14 11:00 .font-unix/
-rw-rw-r-- 1 sunith sunith 0 Aug 14 12:14 Horizon.txt
drwxrwxrwt 2 root root 4096 Aug 14 11:00 .ICE-unix/
drwx----- 4 root root 4096 Aug 14 11:55 snap-private-tmp/
```

Conclusion : Successfully implementation is completed

10. Rename a file.

Objective : Rename a file

Purpose : Changing the name of an existing file to a new name

Implementation :

```
sunith@sunith-virtual-machine:~$ touch file1
sunith@sunith-virtual-machine:~$ ls
file1 Screenshots snap
sunith@sunith-virtual-machine:~$ mv file1 techie
sunith@sunith-virtual-machine:~$ ls
Screenshots snap techie
sunith@sunith-virtual-machine:~$
```

Conclusion : Successfully Implementation is Completed

11. Delete a file.

Objective : Delete a file

Purpose : Removing it permanently from our server

Implementation :

```
sunith@sunith-virtual-machine:~$ mv file1 techie
sunith@sunith-virtual-machine:~$ ls
Screenshots snap techie
sunith@sunith-virtual-machine:~$ rm techie
sunith@sunith-virtual-machine:~$ ls
Screenshots snap
sunith@sunith-virtual-machine:~$
```

Conclusion : Successfully implementation is completed

12. Grant or revoke permissions on a file or directory.

Objective : Grant or revoke permissions on a file or directory

Purpose : Means changing who is allowed to read, write (edit), or execute (run) a file or folder

Implementation :

```
~$ ls -l techie.txt
-rw-rw-r-- 1 sunith sunith 0 Aug 14 12:35 techie.txt
~$ ls -l .viminfo
-rw-r--r-- 1 sunith sunith 1188 Aug 14 11:54 .viminfo
```

Conclusion : Successfully implementation is Completed

13. View the current date and time.

Objective : View the current date and time

Purpose : To check the date and time

Implementation :

```
sunith@sunith-virtual-machine:~$ date
Friday 14 August 2026 12:38:35 PM IST
sunith@sunith-virtual-machine:~$
```

Conclusion : Successfully Implementation is Completed

14. Check the system uptime.

Objective : Check the system uptime.

Purpose : Finding out exactly how long your server has been running continuously without being shut down or restarted

Implementation :

```
sunith@sunith-virtual-machine:~$ uptime
12:39:33 up 1:38, 6 users, load average: 0.00, 0.03, 0.01
sunith@sunith-virtual-machine:~$ uptime -p
up 1 hour, 38 minutes
sunith@sunith-virtual-machine:~$
```

Conclusion : Successfully Implementation is completed

15. View the running processes.

Objective : View the running processes.

Purpose : Asking the server to show you a live list of every program, application, and background task that is currently active and using the system's memory and CPU

Implementation :

```
top - 12:45:22 up 1:43, 6 users, load average: 0.00, 0.00, 0.00
Tasks: 321 total, 1 running, 320 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.5 us, 0.5 sy, 0.0 ni, 98.8 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 2878.4 total, 170.7 free, 1134.9 used, 1572.8 buff/cache
MiB Swap: 2140.0 total, 2139.0 free, 1.0 used. 1441.7 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1751	sunith	20	0	4309160	261100	122404	S	1.0	8.9	1:17.82	gnome-shell
3208	root	20	0	545020	166572	37604	S	0.0	5.7	0:04.06	fwupd
2296	sunith	20	0	235344	79948	54780	S	0.0	2.7	0:01.07	Xwayland
2370	sunith	20	0	547508	77264	58376	S	0.0	2.6	0:00.34	gsd-xsettings
2048	sunith	20	0	837132	61952	46340	S	0.0	2.1	0:00.42	evolution-alarm
4821	sunith	20	0	2787152	61152	44316	S	0.0	2.1	0:00.80	gjs
2464	sunith	20	0	836548	57772	43020	S	0.0	2.0	0:00.45	gnome-calendar
2469	sunith	20	0	554308	52056	39624	S	0.0	1.8	0:15.92	gnome-terminal-
796	root	20	0	1327504	40840	25472	S	0.0	1.4	0:02.27	snappd

Conclusion : Successfully Implementation is Completed

16. Kill a running process.

Objective : Kill a running process

Purpose : Forcing an active program, application, or background task to stop and close immediately.

Implementation

```
sunith@sunith-virtual-machine:~$ sleep 300 &
[1] 4943
sunith@sunith-virtual-machine:~$ ps aux | grep sleep
sunith      4943  0.0  0.0  8372 2064 pts/5    S   12:47   0:00  sleep 300
sunith      4945  0.0  0.0  9080 2644 pts/5    S+  12:47   0:00  grep --color=auto  sleep
sunith@sunith-virtual-machine:~$ id sleep
id: 'sleep': no such user
sunith@sunith-virtual-machine:~$ kill 4943
sunith@sunith-virtual-machine:~$ ps aux | grep sleep
sunith      4948  0.0  0.0  9080 2644 pts/5    S+  12:49   0:00  grep --color=auto  sleep
[1]+  Terminated                  sleep 300
sunith@sunith-virtual-machine:~$
```

Conclusion : Successfully Implementation Is Completed

17. Install a package using the package manager (e.g., apt or yum).

Objective : Install a package using the package manager (e.g., apt or yum)

Purpose : Downloading and installing the software application

Implementation :

```
sunith@sunith-virtual-machine:~$ sudo apt install git
[sudo] password for sunith:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.34.1-1ubuntu1.17).
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
sunith@sunith-virtual-machine:~$
```

Conclusion : Successfully implementation is Completed

18. Update the system packages.

Objective : Update the system packages.

Purpose : Refreshing your server's software catalog and upgrading all installed applications, security patches

Implementation :

```
sunith@sunith-virtual-machine:~$ sudo apt install git
[sudo] password for sunith:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.34.1-1ubuntu1.17).
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
sunith@sunith-virtual-machine:~$
```

Conclusion : Successfully Implementation is Completed

19. Create a symbolic link.

Objective : Create a symbolic link.

Purpose : Creating a special shortcut file that points directly to another file or folder on your system

Implementation :

```
sunith@sunith-virtual-machine:~$ ls
sunith@sunith-virtual-machine:~$ echo "This is my main line" >> techie.txt
sunith@sunith-virtual-machine:~$ ls
techie.txt
sunith@sunith-virtual-machine:~$ ln -s techie.txt symbolic_link.txt
sunith@sunith-virtual-machine:~$ ls
symbolic_link.txt  techie.txt
sunith@sunith-virtual-machine:~$ cat symbolic_link.txt
This is my main line
sunith@sunith-virtual-machine:~$ cat techie.txt
This is my main line
sunith@sunith-virtual-machine:~$
```

Conclusion : Successfully Implementation Is Completed

20. Search for files using the find command.

Objective : Search for files using the find command

Purpose : Telling the server to scan through your folders to hunt down specific files or directories based on rules you choose

Implementation :

```
sunith@sunith-virtual-machine:~$ sudo find /tmp -iname techie.txt
/tmp/techie.txt
sunith@sunith-virtual-machine:~$
```

Conclusion : Successfully Implementation Is completed

21. Compress and decompress files using tar.

Objective : Compress and decompress files using tar

Purpose : Packing multiple files and folders together into a single file to make them smaller and easier to move.

Implementation :

```
sunith@sunith-virtual-machine:~$ mkdir techie
sunith@sunith-virtual-machine:~$ ls
techie
sunith@sunith-virtual-machine:~$ tar -czvf archive.tar techie
techie/
sunith@sunith-virtual-machine:~$ ls
archive.tar  techie
sunith@sunith-virtual-machine:~$ rmdir techie
sunith@sunith-virtual-machine:~$ ls
archive.tar
sunith@sunith-virtual-machine:~$ tar -xzvf archive.tar
techie/
sunith@sunith-virtual-machine:~$ ls
archive.tar  techie
sunith@sunith-virtual-machine:~$
```

Conclusion : Successfully Implementation is completed

22. Monitor system resources with top or htop.

Objective : Monitor system resources with top or htop

Purpose : To check the CPU Utilization

Implementation :

```
sunith@sunith-virtual-machine:~$ ps aux
USER          PID  %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1   0.0  0.4 166840 12112 ?        Ss   11:51   0:04 /sbin/init auto noprompt splash
root           2   0.0  0.0      0     0 ?        S    11:51   0:00 [kthreadd]
root           3   0.0  0.0      0     0 ?        S    11:51   0:00 [pool_workqueue_release]
root           4   0.0  0.0      0     0 ?        I<   11:51   0:00 [kworker/R-rcu_g]
root           5   0.0  0.0      0     0 ?        I<   11:51   0:00 [kworker/R-rcu_p]
root           6   0.0  0.0      0     0 ?        I<   11:51   0:00 [kworker/R-slub_]
root           7   0.0  0.0      0     0 ?        I<   11:51   0:00 [kworker/R-netns]
root          12   0.0  0.0      0     0 ?        I<   11:51   0:00 [kworker/R-mm_pe]
root          13   0.0  0.0      0     0 ?        I    11:51   0:00 [rcu_tasks_kthread]
root          14   0.0  0.0      0     0 ?        I    11:51   0:00 [rcu_tasks_rude_kthread]
root          15   0.0  0.0      0     0 ?        I    11:51   0:00 [rcu_tasks_trace_kthread]
root          16   0.0  0.0      0     0 ?        S    11:51   0:00 [ksoftirqd/0]
root          17   0.0  0.0      0     0 ?        I    11:51   0:01 [rcu_preempt]
root          18   0.0  0.0      0     0 ?        S    11:51   0:00 [migration/0]
root          19   0.0  0.0      0     0 ?        S    11:51   0:00 [idle_inject/0]
root          20   0.0  0.0      0     0 ?        S    11:51   0:00 [cpuhp/0]
root          21   0.0  0.0      0     0 ?        S    11:51   0:00 [cpuhp/1]
```

Conclusion : Successfully Implementation is Completed.